

HOMEWORK 7

1. Let $\mathcal{C}$ be a category. Consider the category $Ar(\mathcal{C})$ of all diagrams of the shape $\bullet \rightarrow \bullet$. Determine all initial and final objects in $Ar(\mathbf{Sets})$.
2. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be two categories with initial objects A_1 and A_2 respectively. Prove that (A_1, A_2) is an initial object in $\mathcal{C}_1 \times \mathcal{C}_2$.
3. Give an example of a category $\mathcal{C}$ and a full subcategory $\mathcal{C}'$ such that the initial objects of $\mathcal{C}$ and $\mathcal{C}'$ are different.
4. A morphism $f : A \rightarrow B$ in a category $\mathcal{C}$ is called *monomorphism* if for every two morphisms $g, h : C \rightarrow A$, the equality $fg = fh$ implies $g = h$. Show that the composition of two monomorphisms is a monomorphism. Determine all monomorphisms in $\mathbf{Sets}$ and $\mathbf{Groups}$. Define the dual notion of *epimorphism* and determine all epimorphisms in $\mathbf{Sets}$.
5. An object P of a category $\mathcal{C}$ is called *projective*, if for every epimorphism $f : B \rightarrow C$ and every morphism $g : P \rightarrow C$ there is a morphism $h : P \rightarrow B$ such that $fh = g$. Show that the free groups in $\mathbf{Groups}$ are projective objects. (You may use without proof the fact that epimorphisms in $\mathbf{Groups}$ are surjective homomorphisms.)
6. Let X be an object of a category $\mathcal{C}$. Consider a new category $\mathcal{C}/X$ with objects the morphisms $f : Y \rightarrow X$ for Y in $\mathcal{C}$ and morphisms between $f : Y \rightarrow X$ and $g : Z \rightarrow X$ being morphisms $h : Y \rightarrow Z$ such that $gh = f$. The product of two objects $f : Y \rightarrow X$ and $g : Z \rightarrow X$ in $\mathcal{C}/X$ is called the *fiber product of Y and Z over X* , denoted $Y \times_X Z$. Show that fiber products exist in $\mathcal{C}/X = \mathbf{Sets}/X$ and $\mathbf{Groups}/X$.
7. The *equalizer* of a pair of morphisms $f, g : X \rightarrow Y$ in a category $\mathcal{C}$ is a morphism $h : Z \rightarrow X$ such that $fh = gh$ and for any morphism $i : T \rightarrow X$ in $\mathcal{C}$ such that $fi = gi$ there exists a unique morphism $j : T \rightarrow Z$ such that $hj = i$. Show that the equalizers exist in $\mathbf{Sets}$ and $\mathbf{Groups}$.
8. Let X be a final object of a category $\mathcal{C}$. Prove that $X \times Y \simeq Y$ for every object Y in $\mathcal{C}$.
9. Let G be a group and let $\underline{G}$ be the category with $Ob(\underline{G}) = \{*\}$ and $Mor(*, *) = G$. Is there a product $* \times *$ in $\underline{G}$?
10. Prove that if the product $X \times Y$ is a final object in the category, then X and Y are also final objects.